

DUPIRE'S LOCAL VOLATILITY

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1 Motivation

Dupire (1994, Risk) builds on the observation that volatility implied by the Black and Scholes (1973) model is different across maturities and strikes. In his “Theory of Rational Option Pricing” Merton (1973) solves the first problem by allowing the volatility to be time-dependent – inferred from the term structure of implied volatility. The second problem is more subtle as it is mainly solved with market incompleteness by introducing stochastic volatilities and/or jumps (Merton 1976; Hull and White 1987; Heston 1993; Bates 1996).

Therefore, a local volatility is an attempt to build a simpler one-dimensional model that would match the volatility surface, preserving market completeness¹. More formally, local volatility is the function of time (to match the variability of volatility across maturities) and the underlying asset’s price (to match the “smile”), so that instead of Black and Scholes (1973)

$$dS_t = (r - \delta)S_t dt + \sigma S_t dW_t^{\mathbb{Q}}$$

or Merton (1973)

$$dS_t = (r - \delta)S_t dt + \sigma(t)S_t dW_t^{\mathbb{Q}}$$

the process becomes

$$dS_t = (r - \delta)S_t dt + \sigma(S, t)S_t dW_t^{\mathbb{Q}} \tag{1}$$

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¹Essentially allowing a perfect hedge against the sources of risk present with just an underlying. Clearly, volatility and/or jumps are not traded.

2 General Setup

The preliminaries are the discount factor

$$P(t, T) = e^{-\int_t^T r_u du} \quad (2)$$

Kolmogorov forward (Fokker-Planck) equation for the pdf $f(S, t)$ ²

$$\frac{\partial f}{\partial t} = -\frac{\partial}{\partial S}(r - \delta)S_t f(S, t) + \frac{1}{2} \frac{\partial^2}{\partial S^2} \sigma^2(S, t) S_t^2 f(S, t) \quad (3)$$

and the European call price is

$$\begin{aligned} C_t &= P(t, T) \mathbb{E}_t^\mathbb{Q}[(S_T - K)^+] \equiv P(t, T) \int_K^\infty (S - K) f(S, T) dS \\ &= P(t, T) \left(\int_K^\infty S_T f(S, T) dS - K \int_K^\infty f(S, T) dS \right) \equiv P(t, T) \int_K^\infty S_T f(S, T) dS + K \frac{\partial C}{\partial K} \end{aligned} \quad (4)$$

following from

$$\frac{\partial^2 C}{\partial K^2} = P(t, T) \frac{\partial}{\partial K} \left(- \int_K^\infty f(S, T) dS \right) \equiv P(t, T) \frac{\partial}{\partial K} (F(K) - 1) = P(t, T) f(S, T) \quad (5)$$

$$\begin{aligned} \frac{\partial^2 C}{\partial T^2} &= -rC + P(t, T) \int_K^\infty (S_T - K) \frac{\partial f(S, T)}{\partial T} dS \\ &= -rC + P(t, T) \int_K^\infty (S_T - K) \left(-\frac{\partial}{\partial S}(r - \delta)S_T f(S, T) + \frac{1}{2} \frac{\partial^2}{\partial S^2} \sigma^2(S, T) S_T^2 f(S, T) \right) dS \end{aligned} \quad (6)$$

where the last derivative is obtained with the help of Kolmogorov forward equation (3). There are two integrals to evaluate. The first one is

$$(r - \delta) \int_K^\infty (S_T - K) \frac{\partial}{\partial S} S_T f(S, T) dS$$

which is obtained via integration by parts³ with $u = S_T - K$ and $v = S_T f(S, T)$:

$$-(r - \delta)(S_T - K) S_T f(S, T) \Big|_{S_T=K}^{S_T=\infty} + (r - \delta) \int_K^\infty S_T f(S, T) dS = (r - \delta) \int_K^\infty S_T f(S, T) dS$$

which, from (4), becomes

$$(r - \delta) \left(\frac{C}{P(t, T)} + \frac{K}{P(t, T)} \frac{\partial C}{\partial K} \right)$$

²Given in any textbook as well as wikipedia.

³ $\int u dv = uv - \int v du$.

The second integral

$$\frac{1}{2} \int_K^\infty (S_T - K) \frac{\partial^2}{\partial S^2} \sigma^2(S, T) S^2 f(S, T) dS$$

is also obtained via the integration by parts with $u = S_T - K$ and $v = \frac{\partial}{\partial S}(\sigma^2(S, T) S_T^2 f(S, T))$:

$$\begin{aligned} \frac{1}{2} (S_T - K) \frac{\partial}{\partial S} (\sigma^2(S, T) S_T^2 f(S, T)) \Big|_{S_T=K}^{S_T=\infty} - \frac{1}{2} \int_K^\infty \frac{\partial}{\partial S} (\sigma^2(S, T) S_T^2 f(S, T)) dS \\ = -\sigma^2(S, T) S_T^2 f(S, T) \Big|_{S_T=K}^{S_T=\infty} = \sigma^2(K, T) K_T^2 f(K, T) \end{aligned}$$

which, as in (5), is

$$\frac{\sigma^2(K, T) K_T^2}{P(t, T)} \frac{\partial^2 C}{\partial K^2}$$

3 Derivation

The Dupire's equation (6) is, therefore,

$$\frac{\partial C}{\partial T} = \frac{1}{2} \sigma^2(K, T) K^2 \frac{\partial^2 C}{\partial K^2} + (r - \delta) \left(C - K \frac{\partial C}{\partial K} \right) - rC$$

so that the local volatility for maturity T and strike K is

$$\sigma(K, t) = \sqrt{\frac{\frac{\partial C}{\partial T} + \delta C + (r - \delta) K \frac{\partial C}{\partial K}}{\frac{1}{2} K^2 \frac{\partial^2 C}{\partial K^2}}} \quad (7)$$

The elegance of Dupire's approach lies in its reversal of the traditional modelling paradigm. Instead of postulating a model for the asset and then deriving option prices, Dupire starts from market-observed vanilla option prices – across strikes and maturities – and asks: what dynamics must the underlying asset follow so that those prices are consistent with no-arbitrage? The result is a model-implied local volatility surface (7) that perfectly fits all market vanilla options. The beauty is that this local volatility is not an input but a consequence of market data: a deterministic function extracted from observed option prices via partial derivatives. It produces an arbitrage-free evolution of the underlying that matches the implied volatility surface exactly, while keeping the market complete. This makes it a powerful and tractable tool in practice, especially for pricing and hedging exotic derivatives.

While Dupire's original derivation sets $r = \delta = 0$ for simplicity and this version of the formula we see in textbooks, this assumption is generally not realistic in practice. Interest rates and dividend yields affect both

the forward price and the shape of the option surface. For actual implementation, the full version of Dupire’s formula (7) must be used, with rates derived from market instruments. Moreover, it is important to note that they can also be time-varying, r_t and δ_t (as e.g. (2) allows).

4 Comparison to Stochastic Volatility Models

Here I offer a little discussion the theoretical and practical differences between local volatility and stochastic volatility models, particularly in how they capture the implied volatility surface and their suitability for pricing and hedging vanilla and exotic options. On the theoretical front, local volatility assumes that volatility is a deterministic function of time and the underlying’s price. This ensures a complete market: all risks can, in principle, be hedged with the underlying alone. By construction, the Dupire’s framework exactly reproduces the observed implied volatility surface of vanilla options at a given moment. Stochastic volatility, by contrast, models volatility itself as a random process, introducing an additional source of uncertainty. This creates an incomplete market where volatility risk cannot be perfectly hedged, but it captures the empirical features of volatility dynamics more faithfully. From a practitioner’s point of view, both models can be calibrated to market vanilla options, but in different ways. Local volatility provides an exact fit to the current implied volatility surface, making it attractive for pricing instruments that depend only on the terminal distribution of the underlying. However, its assumption of deterministic volatility often leads to unrealistic dynamics of the implied smile over time, and it may misprice exotic products that are sensitive to the future evolution of volatility. Stochastic volatility models, while typically offering only an approximate fit to the initial surface, better capture the smile dynamics observed in the market and therefore tend to produce more robust prices and hedges for exotic payoffs.

It is also worth noting that any stochastic volatility model implies a corresponding local volatility surface via Dupire’s formula — both models are consistent at the level of terminal distributions. The distinction lies in the path-wise behaviour: local volatility projects the market into a single deterministic volatility world, whereas stochastic volatility acknowledges and models the randomness of volatility itself. In practice, local volatility remains a useful tool for calibration and intuition, while stochastic volatility is often preferred for products and risk management strategies that rely on a realistic treatment of volatility risk and smile evolution. In the presence of stochastic volatility, local volatility is nearly twice as steep as the implied volatility (Derman et al. 1995).